TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Electrochemical Equivalent of Copper

Pass a measured charge through copper sulphate and weigh what it plates out: Faraday's first law reduced to one number — the mass liberated by one coulomb.

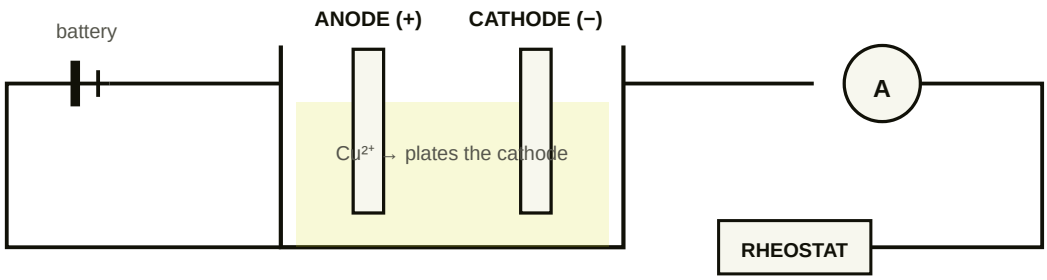
PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 6 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to find the electrochemical equivalent (E.C.E.) Z of copper with a copper voltameter. When current passes through copper sulphate solution between copper electrodes, Cu^{2+} ions discharge at the **cathode** and plate it with fresh copper, while the **anode** dissolves at the same rate, keeping the solution's concentration steady. Faraday's first law of electrolysis states that the mass deposited is proportional to the total charge passed:

$$m = Z\,I\,t \quad \rightarrow \quad Z = m / (I\,t)$$

Z — the mass deposited by one coulomb — is a constant of the ion. For copper it should agree with Faraday's second law: one mole of Cu^{2+} needs 2 moles of electrons (2F of charge, F = 96,485 C), so $Z = M/(2F) = 63.55/(2 \times 96{,}485) \approx 3.29 \times 10^{-4}$ g/C. The experiment weighs the cathode before and after a timed run at a steady, measured current; the anode's matching loss is a built-in check.



COPPER VOLTAMETER — $CuSO_4$ SOLUTION, COPPER PLATES

STEADY CURRENT I FOR A TIMED t · WEIGH THE CATHODE BEFORE AND AFTER

Figure 1 — The copper voltameter in series with ammeter and rheostat. The cathode gains exactly what the anode loses; only the cathode is weighed, cleaned and dried with ritual care.

2. Apparatus and procedure

- **Apparatus:** copper voltameter (glass jar, two copper plates, $CuSO_4$ solution), ammeter, rheostat, battery, key, stopwatch, sensitive balance, emery paper, distilled water, drying arrangement.
- Clean the cathode plate with emery paper, wash, rinse in distilled water, dry and weigh it accurately (m_1). Note which plate is the cathode and mark it unobtrusively.
- Assemble the circuit (Fig. 1), cathode to the negative terminal. Adjust the rheostat to the chosen current (about 1–1.5 A — enough to deposit a weighable mass without rough, powdery plating).
- Start the stopwatch as the key is closed. Keep the current constant throughout (watch the ammeter; trim the rheostat as the battery drifts) for a timed run of about 30 minutes.
- Open the key and stop the watch together. Remove the cathode gently, rinse it in distilled water without rubbing (the deposit must not be wiped away), dry it fully, and weigh again (m_2). Deposit $m = m_2 - m_1$.
- Compute $Z = m/(It)$. If time allows, weigh the anode too — its loss should match m.

3. Worked calculation

Run: I = 1.5 A held constant, t = 30 min = 1,800 s; cathode gains m = 0.889 g.

$$Z = 0.889 / (1.5 \times 1800) = 0.889 / 2700 = 3.29 \times 10^{-4}\ \text{g/C}$$ — in agreement with the theoretical $M/(2F) = 3.29 \times 10^{-4}$ g/C. In SI units $Z = 3.29 \times 10^{-7}$ kg/C.

4. Precautions

- The current density must stay moderate: too high a current plates copper fast and loosely (a spongy deposit that washes off), and side reactions at the electrodes falsify m.
- Current must be truly constant — Z is computed from It, so a drifting current silently averages wrong; log the ammeter reading every few minutes.
- Never rub or touch the plated surface; rinse by dipping, dry completely before weighing (a damp plate weighs the water too).
- Weigh both times on the same balance, with the plate at room temperature; record the anode's loss as a cross-check and investigate any big mismatch.

Practice (viva preparation)

1. State Faraday's first law and define E.C.E. from it.

2. A current of 2 A flows for 45 minutes through a copper voltameter. What mass is deposited (Z = 3.29×10^{-4} g/C)?
3. Why does the anode lose the same mass the cathode gains — and what would happen with inert (platinum) electrodes instead?
4. The cathode is weighed while still slightly damp. In which direction does Z err, and why?
5. From Z alone, how would you estimate the charge on one electron, given Avogadro's number?

Answer key

1. The mass of an ion liberated at an electrode is proportional to the charge passed: $m \propto Q = It$. The E.C.E. Z is the constant of proportionality — the mass liberated by one coulomb.
2. $Q = 2 \times 45 \times 60 = 5{,}400$ C. $m = 3.29 \times 10^{-4} \times 5{,}400 \approx 1.78$ g.
3. For every Cu^{2+} ion discharged at the cathode, a copper atom at the anode gives up two electrons and enters solution as Cu^{2+} — charge conservation keeps the two rates locked, so the solution neither gains nor loses copper overall. With inert electrodes the solution's copper would simply deplete (the anode would evolve oxygen instead), and the blue colour would fade.
4. Z comes out too large: the "gain" weighed includes retained water, inflating m for the same true It.
5. One mole of Cu^{2+} (63.55 g) needs 2F of charge and contains N_A ions, each carrying 2e. Since $Z = M/(2F)$, $F = M/(2Z)$; then $e = F/N_A = 96{,}485/(6.02 \times 10^{23}) \approx 1.60 \times 10^{-19}$ C — electrolysis plus Avogadro's number reaches the electron's charge.
